

Vimal Kamaraj

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Tiruchirappalli / Salem, India

Professional Summary

VLSI Engineer with an M.Tech in VLSI Design and specialized expertise in **Analog Layout** and **Physical Verification**. Proficient in the **Cadence Virtuoso** suite and **Mentor Graphics (Siemens) Calibre** for executing rigorous **DRC/LVS/PEX** sign-off across **45nm to 250nm** CMOS nodes. Proficient in translating complex schematic intent into high-precision silicon through **parasitic-aware routing**, **LDE mitigation**, and optimized **floorplanning**. Technically adept in **VLSI Physical Design** flows, including partitioning and routing, with a proven track record of ensuring signal integrity and area efficiency for first-pass silicon success.

Technical Skills

EDA Toolsets:	Cadence Virtuoso (XL), Calibre, Tanner EDA (L-Edit), Magic, KLayout
Verification:	DRC, LVS, ERC, Antenna Checks, Parasitic Extraction (PEX)
Layout Techniques:	Common-Centroid, Interdigititation, Symmetry Matching, Guard Rings
Process Nodes:	45nm, 90nm, 180nm, 250nm CMOS (GPDK)
Analog Blocks:	Op-Amps, Current Mirrors, Bandgap References, CMOS Logic, CVSL
Programming/OS:	SPICE, Verilog/VHDL, Python, perl, Linux

Projects

High gain Op-Amp

Feb 2026 – March 2026

Analog Layout Design | 45nm / 90nm CMOS

Cadence Virtuoso

- Developed high-precision layout utilizing.
- Optimized **two-stage floorplanning** and device placement to minimize **parasitic coupling** and cross-talk.
- Achieved **72 dB gain**, 2.2 MHz UGB, and 31 μ W power via rigorous **post-layout verification**.
- Executed **DRC/LVS/PEX** sign-off and parasitic-aware analysis for first-pass silicon reliability.

Precision Current Mirror Array

Jan 2026 – Feb 2026

Analog Layout Design | 180nm / 250nm CMOS

Tanner EDA

- Designed **CMOS current mirror circuits** with optimized W/L ratios for accurate current replication.
- Performed **DRC/LVS/PEX verification** and analyzed non-idealities (channel length modulation, output resistance) to improve current stability.

Standard Cell Design & Verification

Nov 2025 – Dec 2025

Full Custom Physical Design | 180nm CMOS

KLayout / Custom Flow

- Developed full-custom layouts for standard logic cells (NAND, Full Adder) optimized for area and speed.
- Conducted **GDSII inspection** and verification using KLayout to ensure mask integrity.
- Implemented **CVSL logic** layouts for high-speed differential signal paths.

Education

Indian Institute of Information Technology IIIT

Tiruchirappalli, TN

M.Tech in VLSI Engineering | **CGPA: 8.55 / 10**

2025 – 2027

Dhirajlal Gandhi College of Technology

Salem, TN

B.E in Electronics and Communication Engineering in Specialization in Fintech and Block Chain | **CGPA: 8.35 / 10** 2021 – 2025

Achievements & Leadership

- Viksit Bharat Delegate:** National dialogue on India Semiconductor Mission. Jan – 2026
- NPTEL Certified:** VLSI Physical Design (Partitioning, Floorplanning, Routing). May – 2026
- GATE Qualified:** Electronics and Communication Engineering Score 342. May – 2026
- OPPO Gen-Green:** Top 100 Intern for sustainable hardware initiatives. Aug – 2025
- Best Academic Performance:** Excellence in Overall academic's. Nov – 2024
- Engineers' Day:** 2nd Prize for precision analog circuit implementation. Nov – 2023